Problem 1

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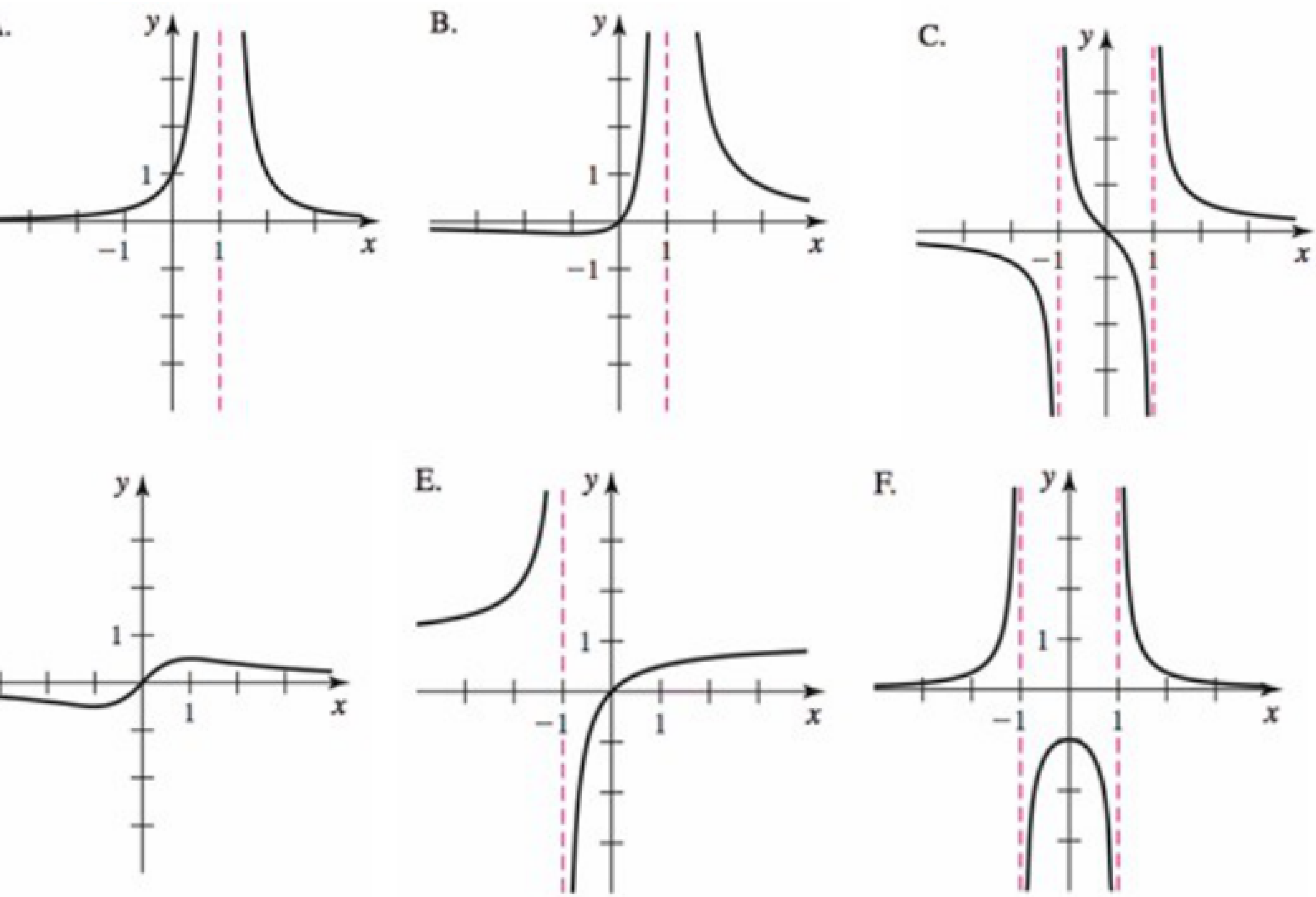
Problem 1 *Without using a graphing utility, match each graph of functions in A-F with the algebraic representation of functions in a-f:*


A.
B.
C.
E.
F.

(a) *The function f defined by* $f(x) = \dfrac{x}{x^2+1}$.

Explanation. Since there is no real number x such that $x^2 + 1 = 0$, we have no candidates for vertical asymptotes. Hence the graph of f should not contain any vertical asymptotes. The only listed graph with no vertical asymptote is graph D.

(b) *The function g defined by* $g(x) = \dfrac{x}{x^2-1}$.

Explanation. *Candidates* for vertical asymptotes:

$$x^2 - 1 = 0 \implies x = 1 \text{ or } x = -1.$$

Test of candidate $x = 1$:

$$\lim_{x\to 1^+} \frac{x}{x^2-1} = \lim_{x\to 1^+} \frac{x}{(x-1)(x+1)} = \infty$$

Note: the limit is of the form $\dfrac{\#}{0}$, the numerator positive, the denominator also positive.
Therefore, vertical asymptote at $x = 1$.

Test of candidate $x = -1$:

$$\lim_{x\to -1^-} \frac{x}{x^2-1} = \lim_{x\to -1^-} \frac{x}{(x-1)(x+1)} = -\infty$$

Note: the limit is of the form $\dfrac{\#}{0}$, the numerator negative, the denominator positive.
Therefore, vertical asymptote at $x = -1$.

Since $g(0) = 0$, the only listed graph that can match is graph C.

(c) *The function h defined by* $h(x) = \dfrac{1}{x^2-1}$.

Explanation. *Candidates* for vertical asymptotes: $x = 1$ and $x = -1$.

Test of candidate $x = 1$:

$$\lim_{x\to 1^+} \frac{1}{x^2-1} = \lim_{x\to 1^+} \frac{1}{(x-1)(x+1)} = \infty$$

Note: the limit is of the form $\dfrac{\#}{0}$, the numerator positive, the denominator positive.
Therefore, vertical asymptote at $x = 1$.

Test of candidate $x = -1$:

$$\lim_{x \to -1^-} \frac{1}{x^2 - 1} = \lim_{x \to -1^-} \frac{1}{(x-1)(x+1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.
Therefore, vertical asymptote at $x = -1$.

Since $h(0) = -1$, the only listed graph that can match is graph F.

(d) *The function a defined by* $a(x) = \dfrac{x}{(x-1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \to 1^+} \frac{x}{(x-1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.
Therefore, vertical asymptote at $x = 1$.

Since $a(0) = 0$ the graph is graph B.

(e) *The function s defined by* $s(x) = \dfrac{1}{(x-1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \to 1^+} \frac{1}{(x-1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.
Therefore, vertical asymptote at $x = 1$.

Since $s(0) = 1$ the graph is graph A.

(f) *The function r defined by* $r(x) = \dfrac{x}{x+1}$.

Explanation. Candidate for the vertical asymptote: $x = -1$.
Test of candidate $x = -1$:

$$\lim_{x \to -1^+} \frac{x}{x+1} = -\infty$$

Note: the limit is of the form $\dfrac{\#}{0}$, the numerator negative, the denominator positive.
Therefore, vertical asymptote at $x = -1$.
Since $r(0) = 0$ the graph is graph E.